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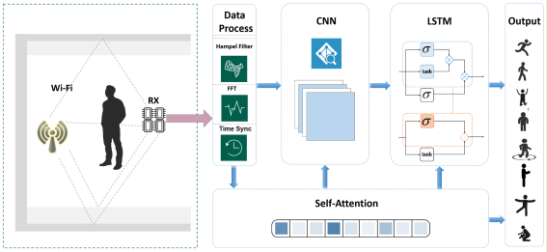
Edge-Enabled Human Activity Recognition Using Hybrid Deep Learning and Multi-Sensor Wi-Fi CSI Arrays

Jiale Wang, *Member IEEE*, Shengmao Que, Ming Xia, Deyou Zhang, Chuang Shi

***Abstract*—Recent advances in wireless sensing have positioned Wi-Fi Channel State Information (CSI) as a promising modality for human activity recognition (HAR), offering a cost-effective and non-intrusive alternative to traditional vision- and wearable-based systems. However, existing CSI-based HAR approaches suffer from high signal volatility, limited feature representation capacity, and poor adaptability in occluded or dynamic envi-ronments—particularly in edge or low-altitude IoT deployment scenarios. To overcome these challenges, we propose an edge-enabled HAR system based on a 2×2 array of Wi-Fi CSI sensors, which synchronously capture multi-channel signals to enhance sensing robustness and spatial diversity. Sensor data is processed locally at the edge via a compact, on-device deep learning framework that integrates convolutional, recurrent, and attention mechanisms. This hybrid architecture is specifically optimized for extracting both spatial and temporal motion features from noisy and rich scattering environments, improving recognition accuracy while reducing transmission and computation overhead. Experiments under both line-of-sight and non-line-of-sight conditions demonstrate the proposed system achieves recognition accuracies of 99% and 97%, respectively, outperforming conventional single-receiver models and demonstrating strong robustness in complex environments. By tightly coupling CSI-based wireless sensing with edge AI, our approach represents a robust and scalable solution for next-generation HAR in complex, low-latency, and privacy-sensitive IoT environments.**



***Index Terms*—Edge Intelligence, Wireless Sensing, Wi-Fi CSI, Human Activity Recognition, Hybrid Deep Learning**

I. Introduction

HAR is a key technology in fields such as navigation, smart healthcare, and health monitoring, with wide-ranging applications. Traditional motion recognition methods rely on various signal sources, such as optical and electrical signals, to identify activities by analyzing obstacles and feature changes during transmission. Early research mainly focused on human behavior analysis and motion recognition, often using wearable sensors to collect acceleration data from different body positions and feeding the data into machine learning (ML) models for decision-making [1]. Additionally, analyzing physiological signals such as electrocardiogram and photoplethysmogram has also been used for activity recognition.

However, the use of wearable sensors in practical applications still faces several challenges. Firstly, most systems require users to wear specialized sensor devices, which limits user freedom of movement and poses inconvenience during wearing and operation. Secondly, traditional methods use limited sensor types and single data types, making it difficult to monitor various human motion features accurately and in real-time. Despite some gait recognition methods such as millimeter-wave radar [2], 5G technology [3], and wearable devices [4], these methods have low adoption rates, and obtaining multi-dimensional features remains a technical challenge. Furthermore, although vision-based recognition methods can capture human activities, they suffer from issues such as blind spots and high energy consumption. In contrast, Wi-Fi CSI, as a monitoring method that does not require wearable devices, has become one of the key technologies for HAR due to its broad coverage, low cost, and real-time data acquisition advantages [5].

Currently, Wi-Fi CSI-based HAR methods face the following main challenges: 1) The data from a single Wi-Fi receiver tends to fluctuate significantly with high noise, often losing essential hidden features, which limits data collection accuracy and affects recognition performance [6]; 2) Traditional machine learning models have limited feature extraction capabilities,

This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFB3910103, and in part by the Fundamental Research Funds for the Central Universities under Grant 501QYJC2025133002. *(Corresponding author: Ming Xia, Deyou Zhang).*

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making it difficult to address both long-term and short-term data characteristics, and therefore are unable to deeply explore the spatiotemporal information in the data [7]; 3) Most existing methods are limited to recognizing simple stationary actions and do not consider applications in wall-penetrating or non-line-of-sight (NLOS) environments [8]; 4) Phase shifts, frequency shifts, and other deviations in Wi-Fi CSI data affect the accuracy, restricting the effectiveness of ML models [9].

To overcome these challenges, this paper exploits CSIs from multiple WiFi sensors and proposes a novel deep learning (DL) framework for HAR. We first developed a 2×2 array Wi-Fi CSI receiver that can synchronously record and merge data from all Wi-Fi channels within the coverage area of the Wi-Fi signal, significantly improving the stability and reliability of the data. Additionally, we incorporated edge computing by integrating a processing unit equipped with an on-chip deep learning model. This unit, implemented on a microcontroller (MCU), processes the CSI data locally, reducing latency and dependence on remote servers.

The deep learning model that deployed on the MCU has integrated CNN, LSTM, and attention mechanisms, which can simultaneously capture short-term and long-term motion features, thereby improving recognition accuracy. The CNN component extracts spatial features from the CSI data, while the LSTM unit captures temporal dependencies, and the Attention mechanism helps focus on the most relevant motion patterns. This design enables effective activity recognition with high precision.

The main contributions of this paper are as follows:

- We propose a system based on a 2×2 array Wi-Fi CSI receiver that can scan and merge all Wi-Fi channel data within area of interest. Compared to a single Wi-Fi receiver, the integration of multi-channel data enhances data stability, leading to improvements across multiple quantitative metrics, including Standard Deviation, Entropy, ADF Test, Volatility, and Mean Absolute Error.
- We design a hybrid CNN-LSTM-Attention (C-L-A)model deployable on a microcontroller for edge computing. This model captures short-term and long-term motion features, extracting spatiotemporal information and focusing on key motion patterns. Edge deployment minimizes latency, reduces reliance on remote servers, and improves real-time performance, making the system more scalable and efficient in practical applications.
- The developed system is capable of operating effectively not only in LOS environments but also in NLOS scenarios with wall penetration, achieving a recognition accuracy of 97%. This enhances the system's robustness and applicability in real-world, complex environments.

The structure of the rest of this paper is as follows. Section 2 reviews the related research published in recent years and discusses their limitations. Section 3 provides a detailed description of the algorithm framework and its implementation steps. In Section 4, experiments and results demonstrate the performance of the proposed seamless positioning method in indoor and outdoor environments. Finally, Section 5 concludes the paper and discusses future work.

II. Related Work

HAR research involves recording and analyzing individuals' daily activities to better understand human behavior. As an important technology in fields such as smart homes, healthcare, and virtual reality, HAR has become a hot research topic [10]. In different application scenarios, the methods of HAR vary. Current mainstream methods typically rely on pure visual models and large dynamic recognition networks. However, due to poor performance in low-light environments, recognition methods based on other signals have gained increasing attention.

To address the limitations of pure visual methods, motion posture recognition based on other signals has emerged. Initially, sensor-based dynamic recognition applications gradually developed, especially the use of Inertial Measurement Units (IMU) in HAR. IMUs do not rely on external information but perceive the acceleration and angular velocity of objects to obtain the status of various body parts and extract features [11], thereby enabling motion state recognition. Despite being simple and practical, the use of wearable sensors presents inconveniences, which is why Wi-Fi wireless sensing has gradually become a viable alternative, especially offering a more reliable solution in NLOS and non-IMU-based applications.

Wi-Fi wireless sensing technology uses received signal strength indicator (RSSI) or CSI to detect human motions. RSSI, as a comprehensive indicator of signal strength, is of-ten used for simple distance estimation [12], whereas CSI reflects more detailed signal propagation states, providing higher precision and more information [13].

A. Existing Human Activity Detection Methods

With the development of electronics, sensor portability, accuracy, and data availability have significantly improved, offering more opportunities for human behavior and health monitoring. From fitness trackers to advanced medical devices, wearable sensors generate a large amount of data, widely applied in health monitoring, activity recognition, and personalized healthcare. The sensors currently used for human motion pattern recognition include accelerometers, gyroscopes, EEG, and PPG. Saha et al. [15] proposed an activity recognition method based on accelerometers, utilizing residual networks for pre-training static and dynamic activities. While this achieved some success, the method performed poorly in different environments and had long training times.

Additionally, the limitation of wearing accelerometers restricts their widespread use, whereas ECG-based behavior recognition does not face such issues. Farrokhi et al. [16] improved activity recognition methods based on ECG and inertial sensors, discovering that a combination of ECG and gyroscopes on the right arm improved recognition accuracy, but still faced issues in complexity and energy consumption. Wang et al. [17] made innovations in the application of ultra-wideband (UWB) for HAR, utilizing UWB's strong anti-jamming ability and high resolution in the frequency domain, providing an alternative solution for HAR. Vdoviak and Sledevič [18] used thermal image

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data to capture human behavior and recognize daily actions (e.g., drinking, bending, walking) through model training.

Considering that Wi-Fi signals are nearly ubiquitous indoors, Wi-Fi signals have been widely used for human behavior prediction. Initially, the RSSI from Wi-Fi signals was used for motion recognition. Gu et al. [20] matched RSSI fingerprints with human behavior to achieve a recognition accuracy of 72.47%. However, RSSI contains limited information and struggles to achieve high-accuracy motion recognition. In contrast, CSI, due to its ability to describe multipath propagation of signals, provides richer data information and improves motion recognition accuracy. Wang et al. [21] used software-defined radio to broadcast and parse Wi-Fi signals, employing a novel Convolutional Neural Network to achieve 88.13% accuracy in activity recognition and 95.68% in indoor localization. However, these Wi-Fi-based motion recognition methods mainly focus on direct-path Wi-Fi signals and rely on router communication protocols, making universal signal collection difficult. Notably, Wei et al. [22] systematically reviewed RSSI-based location fingerprint methods, emphasizing that RSSI suffers from inherent limitations such as high susceptibility to environmental interference (e.g., multipath fading and obstacles) and limited information density, which restrict its performance in high-precision sensing tasks. Their findings align with our observations that RSSI-based approaches struggle to capture subtle motion-induced signal variations, particularly in complex scenarios involving occlusions or dynamic environments. In contrast, Wi-Fi CSI overcomes these drawbacks by preserving fine-grained multipath propagation details, enabling more robust feature extraction for human activity recognition—especially in non-line-of-sight conditions where RSSI performance degrades significantly [23], [24].

B. Human Activity Detection and Recognition Models

In studies of pedestrian motion pattern recognition based on Wi-Fi CSI signals, the statistical models, data processing methods, and machine learning approaches used vary. Early researchers employed Gradient Boosting Decision Trees (GBDT) for denoising and encoding Wi-Fi signals [25]. Fangxin Wang et al. [26] collected Wi-Fi data through Intel Wi-Fi 5300 cards and used CNN-LSTM models to combine time and spatial feat

ures. Feng et al. [27] improved classification results through methods like SVM and LSTM. Additionally, improved models such as BLSTM-CNN [28] and Transformer networks [29] have emerged, utilizing the strengths of different models to enhance recognition capabilities. However, most studies have not considered wall-penetrating signal issues in NLOS environments, and many models did not use wall-penetrating signal data for training. In real-world environments, Wi-Fi signals are unlikely to always have a direct path. Recurrent Neural Networks (RNN) have also been used for multi-room human presence detection based on Wi-Fi CSI, but they are still limited to detecting human presence only [30].

In summary, despite significant progress in Wi-Fi CSI-based applications, there are still shortcomings, especially in wall-penetrating signal transmission detection. Most studies remain at the level of human presence detection, rather than advancing to detecting human behavior patterns in wall-penetrating transmission. Additionally, many studies use open-source Wi-Fi CSI datasets, which lack representativeness and are difficult to apply in real-world environments. To overcome these issues, this study employs an ESP32-based Wi-Fi CSI module, combining array configurations to avoid CSI data loss and proposes a real-time multi-sensor array Wi-Fi CSI signal HAR method based on the CNN-LSTM-Attention framework to achieve robust HAR in both LOS and NLOS environments.

III. MATERIALS AND METHODS

This paper proposes a multi-sensor array Wi-Fi CSI data acquisition and HAR system. The system consists of three main parts: data acquisition, data processing, and HAR. First, the ESP32 module is used to acquire Wi-Fi CSI information. To avoid data loss, omissions, or errors that may result from using a single module, a 2×2 array four-channel ESP32 module is developed to complete the data collection task by supplementing and merging data from four independent channels. Then, the data is processed through a data processing module for cleaning, anomaly detection, and high-dimensional data formation through frequency domain expansion. In last process, all processed data is transmitted through Bluetooth module from ESP32 to STM32H7, and the received data is input into the CNN-LSTM-Attention-based deep learning framework for HAR. The overall architecture of the system is shown in Fig. 1.

A. Data Reception and Processing of Wi-Fi Array Sensors

During the collection of Wi-Fi CSI signals, due to the use of array-based acquisition, data from multiple channels require time synchronization and noise processing to ensure accuracy and consistency. First, time synchronization algorithms are applied to align the data from different channels, avoiding data distortion or errors caused by timing discrepancies. Next, to improve the signal quality, the Hampel filter algorithm is employed to effectively remove high-frequency noise and outliers from the data. Through these methods, effective signals are clearly extracted, providing a solid foundation for subsequent data analysis and activity recognition. Finally, the data from all channels are merged into a high-dimensional integrated dataset, which serves as high-quality input for deep learning models.

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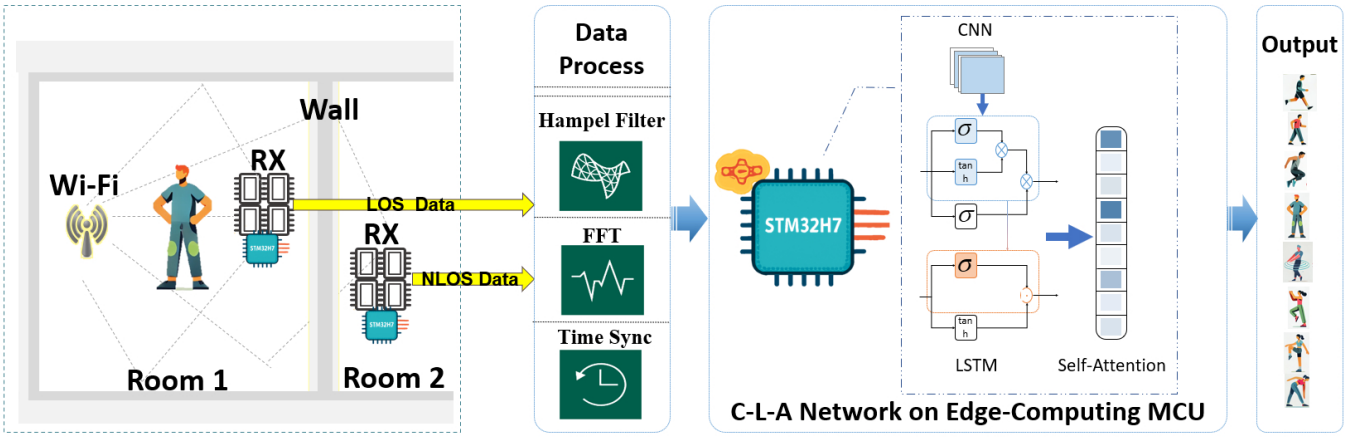

Fig. 1. Human Activity Recognition Framework Using Array-Based Wi-Fi CSI Sensors and Edge-Computing MCU

Wi-Fi CSI describes the attenuation factors along each transmission path, such as signal scattering and decay. Initially, the displacement distance or movement state of a person or object was inferred from changes in simple RSSI signals. The Wi-Fi RSSI signal represents the current strength of a Wi-Fi signal and is expressed as:

$$RSSI = 10log(P_R) \tag{1}$$

where $RSSI$ and P_R represent the received signal power in dB and watts, respectively, and $log(\cdot)$ indicates the logarithmic operation. Typically, due to the relatively weak signal, Wi-Fi $RSSI$ is expressed in dBm. In practice, an RSSI value between 0 and -50 dBm generally indicates a reliable access point, while values below -70 dBm are likely affected by interference. RSSI is an indicator used in wireless communications to measure the strength of the connection between devices and access points, and it can also theoretically be used to estimate the distance between the device and access point. Indoor HAR based on Wi-Fi RSSI has some practical applications. However, since Wi-Fi RSSI signals are a combination of multiple signals, they are easily interfered with by unrelated Wi-Fi signals nearby, and commercial Wi-Fi RSSI detectors have limited accuracy. Therefore, this method is mainly suitable for ideal environments with strong signals and minimal obstructions.

In wireless channel theory, Channel Impulse Response (CIR) is commonly used to evaluate wireless channels. CIR is represented as [31]:

$$h(t) = \sum_{l=1}^{N} \alpha_l e^{-j\beta_l} \delta(t - t_l) \tag{2}$$

where N represents the total number of subcarriers in the signal transmission, δ is the unit impulse function, α_l is the attenuation amplitude at the l th subcarrier, β_l is the phase shift at the l th subcarrier, and t_l is the delay of the l th subcarrier at the current position. From the equation, it can be seen that CSI reflects the fundamental information at the channel level during signal propagation [32]. Thus, each data packet can represent the amplitude and phase of an Orthogonal Frequency Division Multiplexing (OFDM) carrier:

$$H(k) = |H(k)|e^{jH(k)} \tag{3}$$

Clearly, in the above formula, $H(k)$ is the CSI value of the k th channel. Different motion states affect the CSI value with varying amplitudes and time intervals, meaning CSI can accurately describe the attenuation factors along each transmission path and quantify them into a channel gain matrix H. In an OFDM transmission system (Wi-Fi CSI is transmitted via OFDM modulation), the communication of channel state information can be represented as:

$$R = HS + N \tag{4}$$

$$\hat{\varphi}_{H(k)} = \varphi_{H(k)} + 2\pi \frac{H(k)}{N} \tau + \beta + Z \tag{5}$$

where R is the received channel information vector, S is the transmitted channel information vector, H is the equivalent frequency response characteristic of the CSI transmission process, N is the noise signal, τ is the receiver time offset, β is the random phase, and Z is the measurement error [33]. By taking the difference between the phases measured at adjacent carriers, we can obtain:

$$\Delta\hat{\varphi}_{H(\Delta k)} = \Delta\varphi_{H(\Delta k)} + 2\pi \frac{\Delta H(\Delta k)}{N} \tau + \Delta\beta + \Delta Z \tag{6}$$

Based on the above analysis, even with good hardware and configuration, the received CSI data will still be affected by noise and cannot be directly utilized in its raw form. Specifically, CSI data can be influenced by Carrier Frequency Offset, Sampling Frequency Offset, and errors in the Analog-to-Digital Conversion process. To address this, we applied mean filtering and time interpolation methods [34] to handle these noises:

$$y[n] = \frac{1}{N} \sum_{k=0}^{N-1} x[n-k] \tag{6}$$

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$$\begin{cases} S_i(x) = a_i(x-x_i)^3 + b_i(x-x_i)^2 + c_i(x-x_i) \cdot \\ S_i^{'}(x_{i+1}) = S_{i+1}^{'}(x_{i+1}) \\ S_i^{''}(x_{i+1}) = S_{i+1}^{''}(x_{i+1}) \end{cases} \quad (7)$$

Where $i = 0,1,\dots,n-1$, $x[n]$ represents the Wi-Fi CSI input data, and $y[n]$ represents the data after mean filtering. $S_i(x)$ is the i th polynomial function obtained by interpolating and fitting the $x[n]$ data, with S_i' and S_i'' representing the first and second derivatives of S_i. The coefficients a, b, c, and d are determined by solving the given conditions.

Since these noises mainly arise from environmental or interference noise and can be approximately modeled as Gaussian distributed, they can be filtered using the mean of the four Wi-Fi CSI receivers at the same time. Additionally, to achieve time synchronization of the data from the four ESP32 Wi-Fi CSI receivers, we calibrated the local clock of the ESP32 with the clock obtained during Wi-Fi communication, ensuring data synchronization for subsequent interpolation or sorting processing.

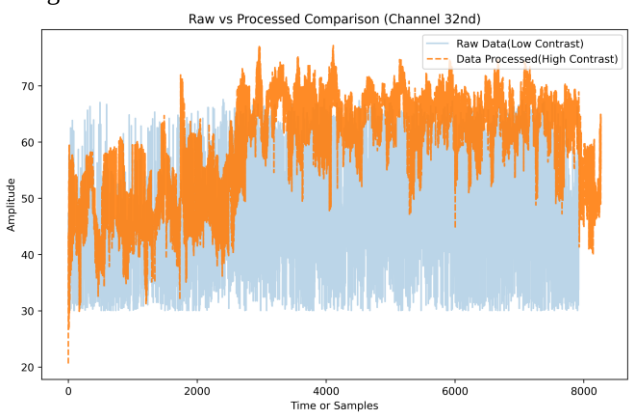

Fig. 2. Comparison of Original and Processed Data

At the same time, for outliers during the sampling process, i.e., the anomalies in the data, we applied the Hampel filter method [35]:

$$M = median(x[n-k],\dots,x[n],\dots,x[n+k]) \quad (8)$$

$$MAD = median(|x[i]-M|),\ \ i \in [n-k, n+k] \quad (9)$$

$$x[n] = \begin{cases} x[n] & |x[n]-M| < \tau \times MAD \\ M & |x[n]-M| > \tau \times MAD \end{cases} \quad (10)$$

where M is the median of the sequence $x[n]$ within a window of k values before and after n, and MAD is the median of the absolute differences between the sequence $x[n]$ and M. The threshold τ is set to 3 in this paper. By calculating the median and median absolute deviation within a specified window, we assess the central tendency and dispersion of the signal. When a value deviates from the median by more than the set threshold, it is considered an outlier and is replaced with the median of the window.

B. Feature Extraction from Raw Data

The Wi-Fi CSI data received through the ESP32 module contains time-domain state information from 64 different channels, including real and imaginary parts. Each data frame consists of 2×64 data points with a transmission frequency of 20 Hz. Due to the high dimensionality of the data, directly using it for recognition tasks would impose a significant computational burden. Therefore, Principal Component Analysis (PCA) [36] was applied for dimensionality reduction to extract key channel information:

$$\begin{cases} \overline{x_j} = \frac{1}{n}\sum_{i=1}^{n} x_{ij}, \qquad X_{ij}' = x_{ij} - \overline{x_j} \\ \Sigma = \frac{1}{n-1} X^T X \end{cases} \quad (11)$$

$$\Sigma v = \lambda v \quad (12)$$

$$Z = XW \quad (13)$$

where x_{ij} represents the CSI reception sequence, i indicates the sequence position generated by time, and j represents the channel position. $\sum$ is the two-dimensional covariance matrix of X, λ is the eigenvalue, v is the eigenvector, and W consists of the eigenvectors corresponding to the k largest eigenvalues. Z is the reduced data.

In PCA, a variance threshold was designed for feature selection, assuming that the principal components vary across different categories, leading to shifts in data that manifest as larger variances. After filtering, lower-dimensional data was obtained, reducing the computational load for the subsequent CNN-LSTM-Attention-based learning network training.

The experiment was designed with CSI data under both wall-penetrating and LOS conditions, and actions including running, walking, jumping, standing still, spinning, clapping, bending, and leg-lifting. After noise filtering and dimensionality reduction, the CSI data requires further analysis. However, a single time-domain data point cannot clearly distinguish the effects of different activities, as actions at different times may produce similar or indistinguishable effects on the Wi-Fi CSI signal. For example, walking and running may generate similar signal variations when passing through the same location. Therefore, we further considered performing a frequency-domain transformation of the raw data.

Wavelet transform is a flexible and effective signal analysis and image processing tool, based on the decomposition of basis functions in both time and frequency domains, allowing for the analysis of specific signal characteristics. In this study, Haar wavelet transform was used to decompose the signal, simplifying the computation while yielding good transformation results. This choice is primarily motivated by two key considerations for our edge-enabled HAR scenario. First, the Haar wavelet features an extremely simple rectangular basis function, whose transformation involves only basic addition and subtraction operations—resulting in significantly lower computational complexity compared to other wavelet families such as Daubechies

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(which require high-order convolution due to their vanishing moments) or Symlet (which prioritize symmetry at the cost of increased arithmetic load). This lightweight property is critical for deployment on resource-constrained edge devices like the STM32H7 microcontroller, ensuring real-time processing without excessive latency. Second, Wi-Fi CSI signals in human activity recognition are dominated by transient features (e.g., abrupt amplitude or phase changes caused by motion onset/termination). The Haar wavelet's high sensitivity to high-frequency discontinuities makes it particularly adept at capturing these critical patterns, whereas Daubechies or Symlet wavelets—optimized for smoothing continuous signals—offer limited advantages for distinguishing such transient CSI variations induced by activities like walking, running, or jumping.

Low-frequency part:

$$c_a(j,k) = \frac{1}{\sqrt{2}}\left(c_a(j+1,2k) + c_a(j+1,2k+1)\right) \tag{14}$$

High-frequency part:

$$c_d(j,k) = \frac{1}{\sqrt{2}}\left(c_a(j+1,2k) - c_a(j+1,2k+1)\right) \tag{15}$$

where $c_a(j,k)$ represents the approximation coefficients (low-frequency) at level j, and $c_d(j,k)$ represents the detail coefficients (high-frequency) at level j. Experimental results validate that wavelet transform [37] can effectively reveal the characteristic behaviors of different actions in the frequency domain (detailed experimental results and conclusions will be presented in Section IV).

To capture the relationship between the current time and past moments, we considered including the previous ten data points during network training. However, in practical tests, due to high dimensionality, the features may become unclear. Therefore, we compressed the data dimensionality by calculating the mean and variance of the data.

C. CNN-LSTM-Attention Network for Human Activity Recognition

For multivariate data prediction, this paper adopts a model based on the CNN-LSTM-Attention framework, which assigns different labels to different motion phases for accurate action recognition and prediction.

In the context of model selection, LSTM is particularly suited for processing sequential data and capturing long-term dependencies, making it well-aligned with the temporal characteristics of Wi-Fi CSI data. Given that human activities exhibit intrinsic temporal continuity and interdependence, LSTM effectively models these sequential patterns, thereby enhancing recognition accuracy.

However, in NLOS environments, increased signal attenuation and noise interference can exacerbate classification errors when relying solely on LSTM. To mitigate this issue, CNN is integrated as a robust feature extractor, enabling the model to first extract high-level representations from raw CSI data across various conditions before sequentially processing them with LSTM. This hybrid approach ensures more effective feature encoding and reduces susceptibility to environmental variations.

Moreover, to dynamically regulate the interaction between CNN and LSTM, an Attention mechanism is incorporated to assign adaptive weights, optimizing feature fusion and enhancing the model's resilience to complex and dynamic environmental conditions. The well-trained C-L-A model can be easily deployed in STM32H7, which offers a high-performance MCU platform capable of efficiently running various neural network models with minimal adaptation.

The designed CNN-LSTM-Attention framework is shown in Fig. 3. First, the data is preprocessed through a normalization layer. The purpose of normalization is to eliminate interference from different magnitudes, facilitating subsequent calculations. Then, the input data is flattened into a 64×2 vector and passed through a dropout layer with a 20% rate to prevent overfitting. After that, the data enters the convolutional layer, where each convolution kernel has a size of 5×1. The output of the convolutional layer continues as the input to the LSTM model. LSTM is a classic type of recurrent neural network (RNN) that has a time-memory effect [38] and can use historical input data to predict the output at the current time. This property is ideal for the memory demands in action recognition tasks and helps improve model performance [39].

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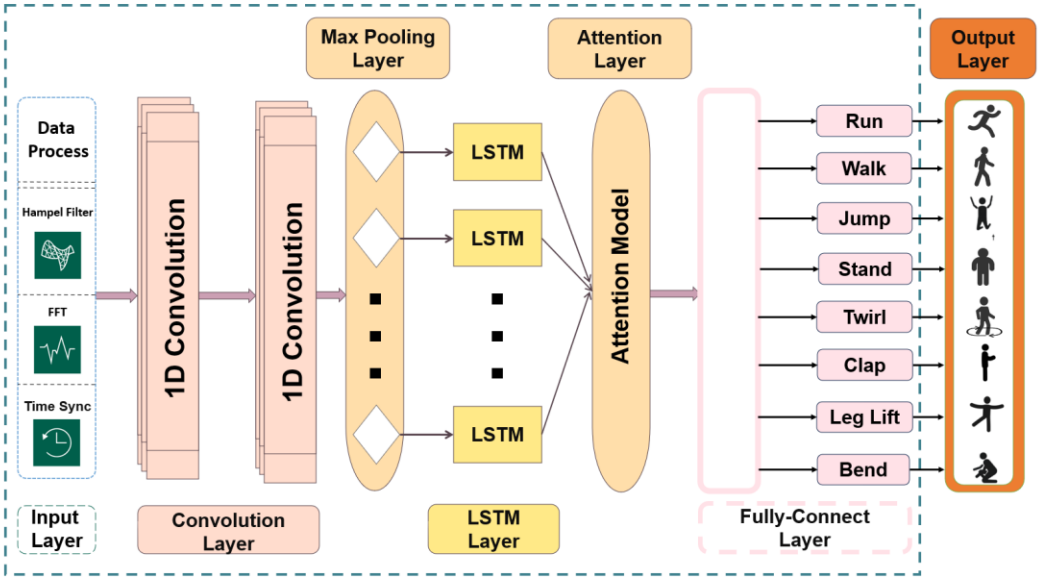

Fig. 3. Hierarchical CNN-LSTM-Attention Network for Human Activity Recognition

Convolutional Layer Operation:

$$X^{(l)} = f\left(W^{(l)} * X^{(l-1)} + b^{(l)}\right) \tag{16}$$

where: $X^{(l)}$: Output feature map from layer, $W^{(l)}$: Convolution kernel weights for layer, $b^{(l)}$:Bias term for layer, $*$: Convolution operation, $f(*)$: Activation function (such as ReLU), where in our experiment, the nonlinear activation function is $f(x) = max(0, x)$.

In LSTM Unit:

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \tag{17}$$

$$f_t = \sigma\left(W_f x_t + U_f h_{t-1} + b_f\right) \tag{18}$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \tag{19}$$

$$\widetilde{c_t} = \tanh(W_c x_t + U_c h_{t-1} + b_c) \tag{20}$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \widetilde{c_t} \tag{21}$$

$$h_t = o_t \odot \tanh(c_t) \tag{22}$$

where: x_t: Input at time step t, h_t: Hidden state at time step t, c_t: Cell state at time step t, i_t: Input gate, f_t: Forget gate, o_t: Output gate, $\widetilde{c_t}$: Candidate cell state; W,U, and b: Weights and bias terms; σ: Sigmoid activation function, $tanh$: Hyperbolic tangent function, $\odot$: Element-wise multiplication.

Attention Mechanism:

$$e_t = v^T tanh(W_a h_t + U_a s_{t-1} + b_a) \tag{23}$$

$$\alpha_t = \frac{\exp(e_t)}{\sum_{k=1}^{T} \exp(e_k)} \tag{24}$$

$$c = \sum_{t=1}^{T} \alpha_t h_t \tag{25}$$

where: v, W_a, U_a, b_a: Learnable parameters in the attention mechanism, α_t: Attention weight at time step t, c: Context vector (the weighted sum of hidden states), s_{t-1} The previous hidden state of the decoder.

Final Output:

$$y = softmax\left(W_y c + b_y\right) \tag{26}$$

where: y: Output category distribution, W_y: Weight matrix for the fully connected layer, b_y: Bias term for the fully connected layer, $softmax(\cdot)$: Softmax function, used to compute the probability distribution over the categories.

In the LSTM, an attention layer is interspersed, applying the Attention strategy to determine which parts of the input data are crucial, allowing the model to focus on these key pieces of information [40].

IV. Experiments and Results

This section presents experimental validation of the model and methods designed in this paper on real-world data. In Section IV-A, we provide detailed descriptions of the experimental setup, including equipment, environment, and content. In Section IV-B, we validate the model using the collected data and perform basic feature analysis, demonstrating improvements in robustness and stability after data fusion processing. In Section IV-C, we showcase the effects of wavelet transformation and Fast Fourier Transform (FFT) on the data, further proving the effectiveness of the data processing. In Section IV-D, we compare the performance parameters of different models during the training process, highlighting the superiority of the model proposed in this paper.

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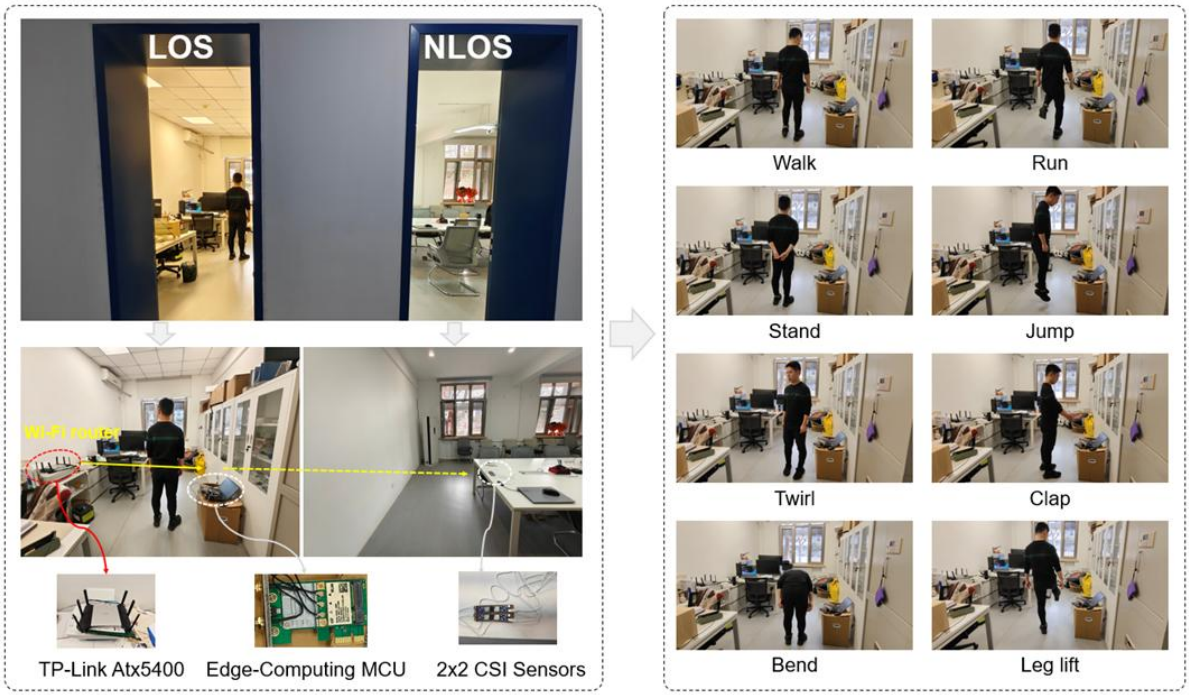

Fig. 4. Experimental equipment, Experimental Setup, and Human Activities in the Experiments.

A. Experimental Setup

In real-world environments, Wi-Fi signals do not always reach the device via direct path; in some environments, signals must pass through walls, forming NLOS reception. To reflect the general applicability in everyday use, we designed two data collection methods: 1) direct LOS Wi-Fi CSI collection and 2) NLOS Wi-Fi CSI collection after passing through walls. Fig. 4 shows the experimental equipment, experimental setup, and various human actions performed during the experiment.

In this experiment, the Wi-Fi signal transmission device is a TP-LINK 5400 router, supporting the Wi-Fi 6 protocol and providing stable signals on both the 5 GHz and 2.4 GHz bands. The Wi-Fi CSI collection chip used is the ESP32-WROOM series from Espressif, which gathers data from surrounding routers' Beacon frames and CSI information. Multiple Access Points continuously broadcast Beacon frames on specific channels to announce their presence. The ESP32 monitors and receives these management frames in sniffing mode to calculate CSI. When a Beacon frame is transmitted on the specific channel that the ESP32 is currently waiting on, it can receive multiple Beacon frames from different APs.

The STM32H7 microcontroller facilitates communication with the ESP32 via Bluetooth, enabling seamless data transfer. The collected CSI data is transmitted from the ESP32 to the STM32H7, which processes the data through a designed C-L-A network. This network performs human action recognition by analyzing the temporal and spatial features of the collected data, identifying different movements such as walking, running, or jumping.

To ensure data stability and integrity, we developed a 2×2 array receiver based on the ESP32 and placed it between the testing subject and the Wi-Fi router. We wrote a serial output program to decode the Wi-Fi CSI information from the ESP32. The 2×2 Wi-Fi receiver array compensates for data loss and errors while effectively avoiding environmental noise interference affecting a single device. However, as shown in the Fig 5, for other configurations such as 3×3 and 4×4 arrays, it is easy to observe that their training duration is longer, and in some cases (3×3 CSI receivers), the accuracy decreases slightly. In addition, larger arrays require more computation and bandwidth, making it difficult to achieve optimal performance on actual hardware.

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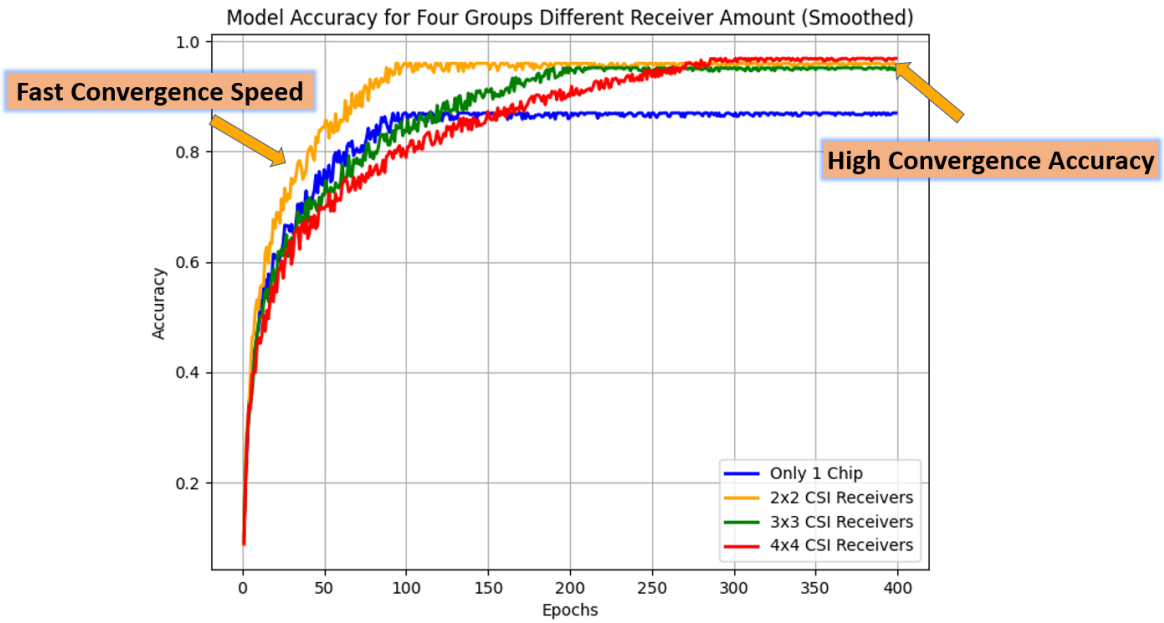

Fig. 5. Accuracy over Epochs for Different Wi-Fi CSI Receiver Configurations.

Unlike conventional multiple-input multiple-output antenna systems, the ESP32-based receiver array offers a lightweight, programmable architecture that enables greater adaptability. Through software configuration, the system can be set to output CSI data exclusively from a specific MAC address or all CSI information received within the current environment. This programmability enhances flexibility and facilitates adaptation to various deployment scenarios.

The experiment was conducted in two independent rooms, 105 and 107, in the East Building of the Library at Beihang University. The Wi-Fi router was placed in room 105, and the two Wi-Fi CSI receiver arrays were placed in rooms 105 and 107, creating LOS and NLOS reception environments. The testing subject performed various actions in room 105, including walking, running, standing still, jumping, turning, clapping, bending, and leg lifting. Each action was sampled for about 90 seconds. The Wi-Fi CSI receiver array's data rate was 20 Hz, effectively capturing the time intervals between different human actions.

Table 1. System Components for Data Collection and Processing

Component	Description
TP-LINK 5400 Router	Wi-Fi 6 router, stable signals on 5 GHz.
2×2 ESP32-based Wi-Fi CSI Receiver Array	ESP32-based array for data collection, compensating for data loss and noise.
STM32H7 Microcontroller	Cortex-M7 microcontroller with up to 480 MHz clock speed, enabling fast computation and low-latency neural network deployment.
Bluetooth Module	Facilitates the wireless transmission of CSI processed by the ESP32 to the STM32H7 microcontroller.
Participant	Person performing various actions like walking, running, jumping, etc., during data collection.

B. Array Wi-Fi Data Feature Analysis

In the face of limitations in serial communication and noise interference, this paper innovatively employs a Wi-Fi CSI receiver array and leverages the complementary nature of the array data to initially remove noise and prevent signal loss. We first conduct a comprehensive analysis of four sets of data collected via direct Wi-Fi signal transmission, processing each set to obtain its amplitude and phase information. As shown in Fig. 6, during the data collection process, a single ESP32 may lose data frames at certain times, represented as black areas in the figure. In other words, parts of the CSI data received by the ESP32 may contain zero or near-zero values, indicating that the channel information is lost during those time intervals.

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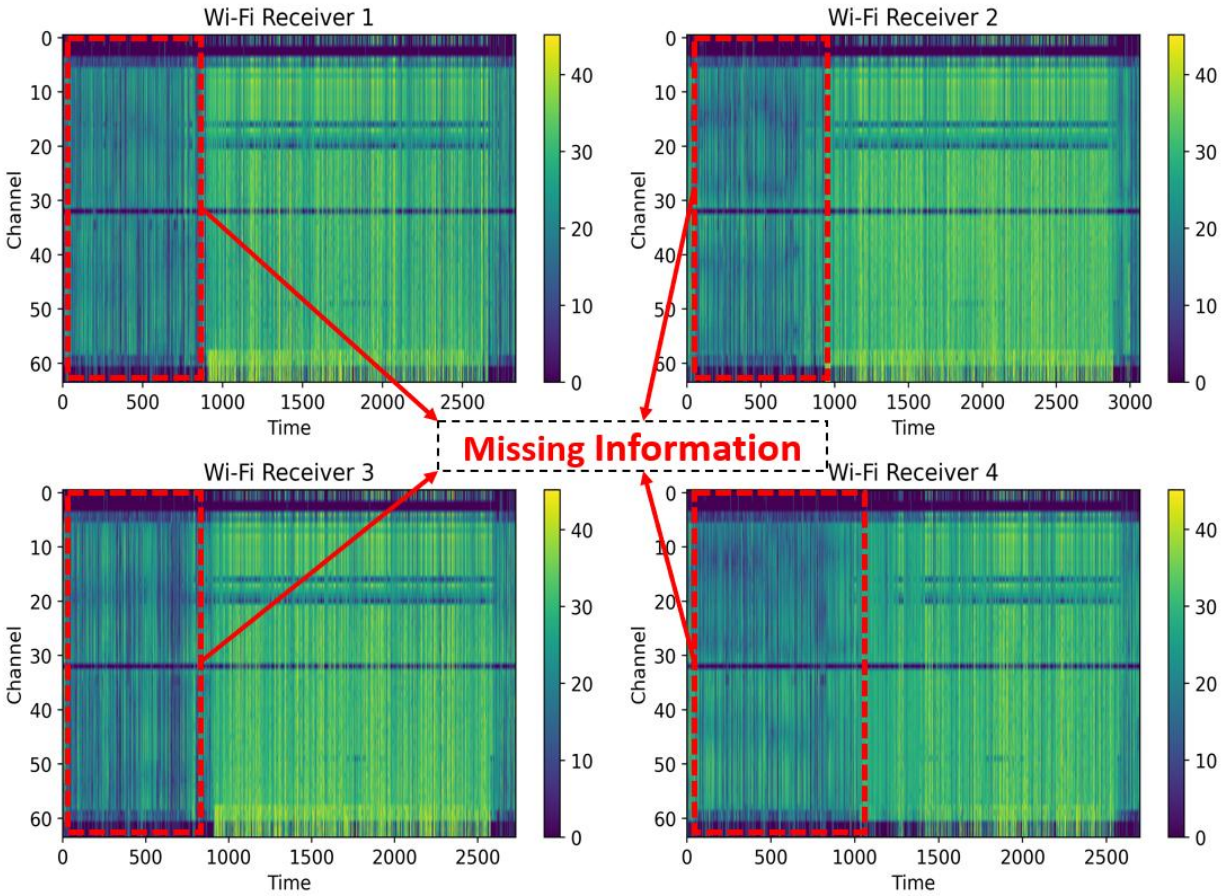

Fig. 6. Array WI-FI CSI Sensor Receiving Data Under LOS Environment Between Router and Receiver

By fusing, interpolating, and averaging the data from four independent receiver channels, the data from the array receiver avoids most data interference, with blank data appearing only at a few points. This significantly enhances the robustness of the data. Fig. 7 also shows that, compared to the single Wi-Fi CSI signal processed similarly, the fused signal contains richer information, has more distinct transition sections, and exhibits better feature distinguishability.

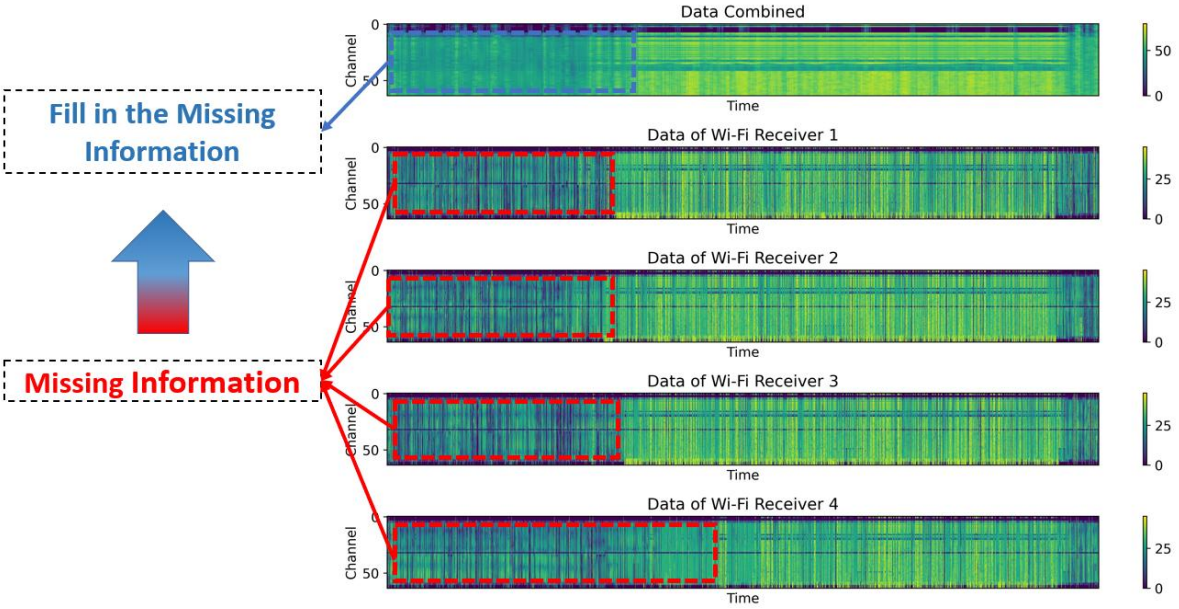

Fig. 7. Wi-Fi CSI signal after amplitude and phase offset compensation in LOS environment.

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For detecting NLOS signals formed when Wi-Fi signals pass through walls, the signal is significantly affected by the wall, making it more challenging to identify distinct feature points and detect differences in actions compared to a line-of-sight (LOS) environment. Although there are still differences in the signals from various receivers, the signals after wall penetration contain more ripple noise compared to those in direct line-of-sight environments.

Furthermore, due to signal obstruction caused by walls, the receiver acquires a reduced amount of CSI data, increasing the likelihood of packet loss. While the CSI data distribution in the NLOS scenario exhibits a general resemblance to that in the LOS scenario, the data length in the NLOS case is significantly shorter. This observation further underscores the impact of NLOS propagation on data integrity and signal attenuation, highlighting potential challenges in CSI-based HAR.

As shown in Fig. 8, we process the Wi-Fi signals and attempt to improve signal quality by compensating for signal amplitude and shifting the phase. The figure clearly shows how the signals change over time, displaying the CSI data from four Wi-Fi receivers on different channels. By further processing these signals, we can partially compensate for signal attenuation caused by environmental factors (such as noise and interference), while phase shift adjustments enhance signal stability and recognizability. After processing the amplitude and phase information, the data exhibits smoother and more consistent features, which aids the accuracy of subsequent action recognition tasks.

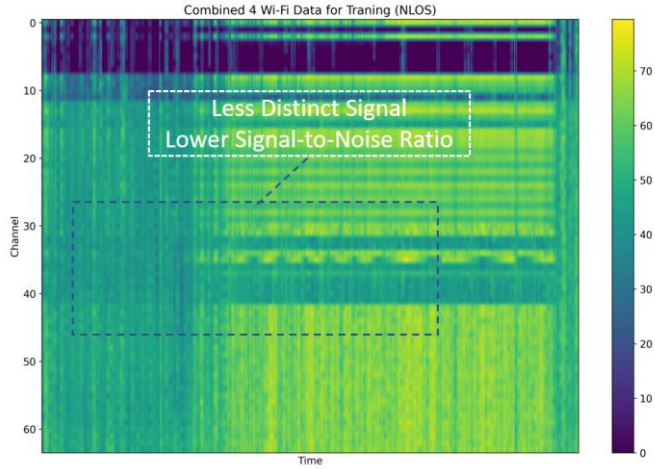
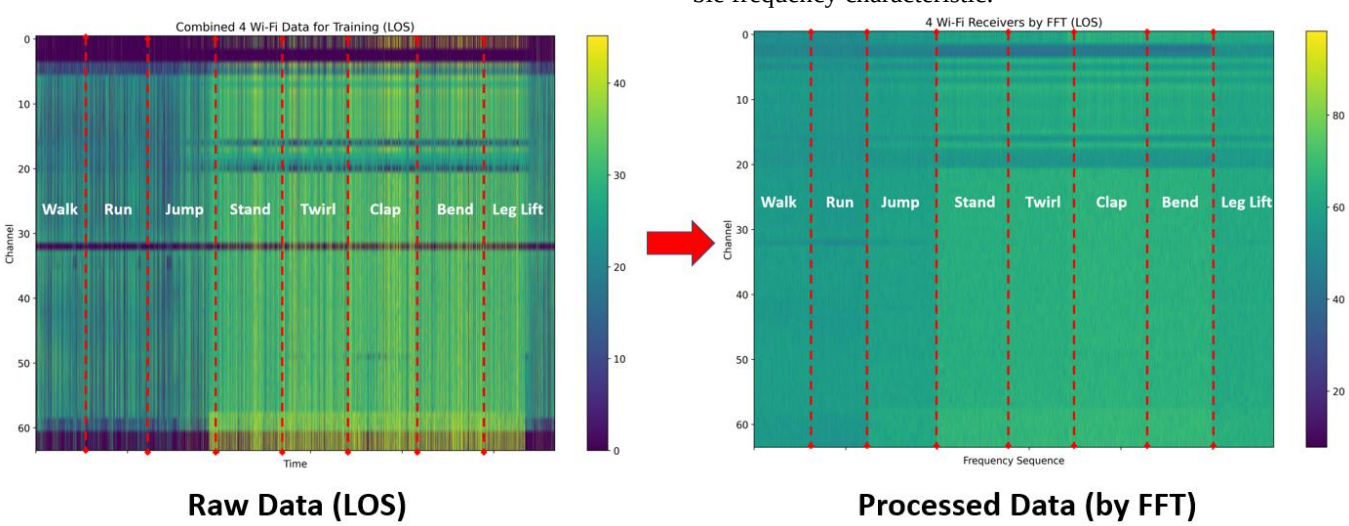


Fig. 8. Wi-Fi CSI signal after amplitude and phase offset compensation in NLOS environment.

C. Data Synchronization and Filtering

From the data figures in the previous section, it is evident that the fused data contains more information and better reflects the characteristics of the behaviors. However, directly inputting the raw data into a neural network for training is not optimal, as this could lead to an excessively lengthy feature search process, extending training time. Therefore, we will now introduce the feature extraction process and its effects.

FFT, as a fast algorithm for discrete Fourier transform, has the characteristic of converting data between time and frequency domains. Since FFT is optimized in many chips and industrial products, we utilize it to convert the time-domain data collected into frequency-domain data. Different actions often have unique characteristics in the frequency domain. Thus, we first apply FFT to the raw sampled data. We use 1-second worth of sampled data (20 time-domain data points) as input to the FFT, and the processed result is shown in Fig. 9(a). From the figure, it is clear that the chaotic data in the time domain becomes more distinct in terms of frequency features after applying the FFT, with each action corresponding to a more noticeable frequency characteristic.



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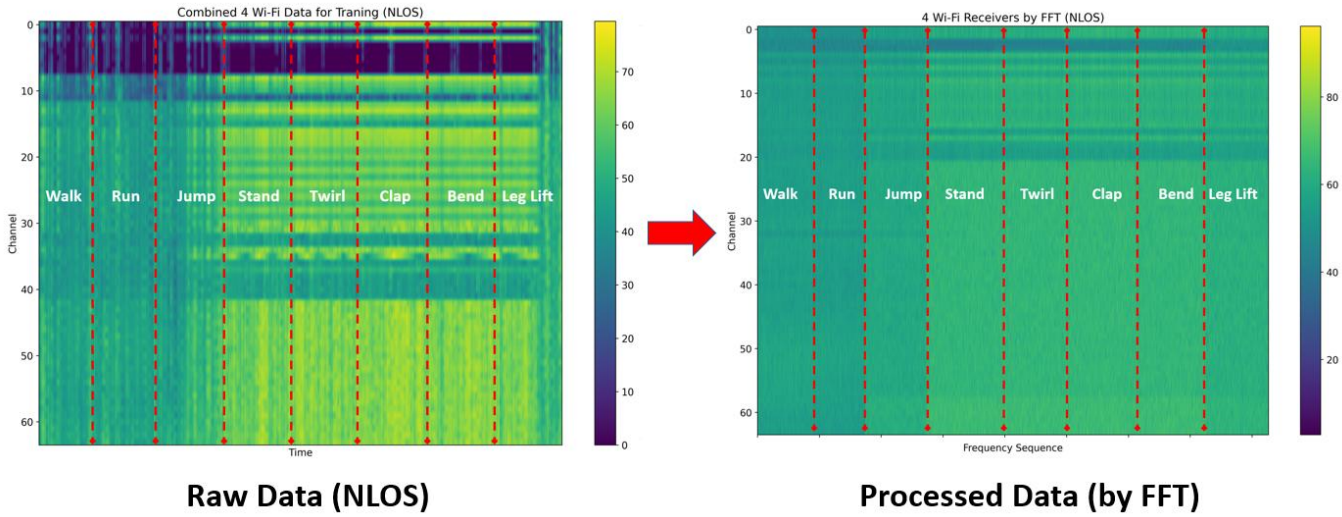
(a) Line-of-Sight environment



(b) Non-Line-of-Sight environment

Fig. 9. FFT-processed result in (a) LOS environments and (b) NLOS environments.

Similarly, we applied FFT to the data collected in the NLOS environment, and the result is shown in Fig. 9(b). In certain action regions (such as standing, turning, clapping, bending over, etc.), the FFT result is difficult to distinguish by visual inspection. Based on this, we hypothesize that further processing is needed, so we consider applying wavelet transform. Wavelet transform, similar to FFT, but with more flexible basis functions, allows for decomposition using various types of orthogonal signals. In this paper, we used the Haar wavelet transform, with its mother wavelet defined as:

$$\psi(t) = \begin{cases} 1 & 0 < t < 1/2 \\ -1 & 1/2 < t < 1 \\ 0 & otherwise \end{cases}$$

And its corresponding scaling function:

$$\phi(t) = \begin{cases} 1 & 0 < t < 1 \\ 0 & otherwise \end{cases} \tag{27}$$

Through wavelet transform, time-domain signals can be analyzed from another perspective. The result of the wavelet transform is shown in Fig. 10, where it can be observed that the wavelet transform offers some degree of discrimination in different regions. Although the results of the wavelet trans-form do not intuitively present motion frequency characteristics like the FFT method, the distribution of high-frequency components within local time segments still provides valuable information, contributing to the improvement of current motion pattern recognition capabilities. Therefore, we com-bine the FFT data with the wavelet transform data as input to the neural network to expand the feature input dimensions and enhance the model's recognition ability.

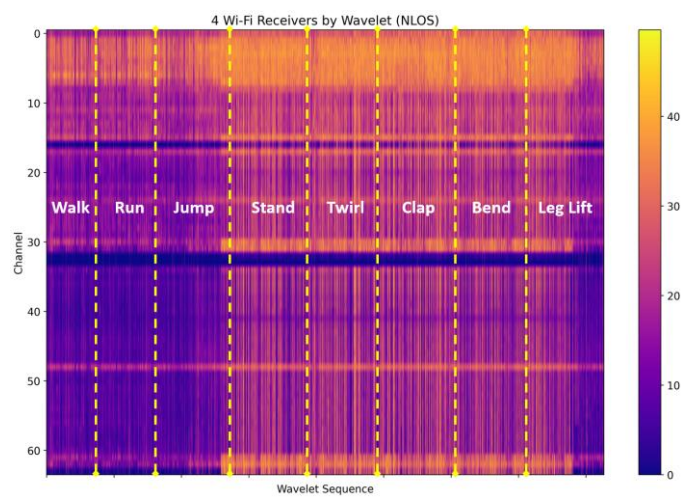


Fig. 10. Wavelet-processed Data in LOS and NLOS Environments

D. Performance Analysis of Action Detection Experiments

After the previous three sections, this paper has completed the cleaning and feature extraction of the raw data. These preliminary processing steps significantly reduce the model training time and improve the model's accuracy. In the experiments, we first compared the performance of a single Wi-Fi receiver with that of the 2×2 array receiver we designed, validating the significant advantage of using the Wi-Fi receiver array over a single receiver in action detection.

We compared the processed data with four individual CSI data sets using multiple data stability evaluation methods, including Standard Deviation, Entropy, ADF Test (ADF_Stat, ADF Test: P_Value), Volatility, and Mean Absolute Error (MAE). As shown in Table. 2, the results indicate that the processed data achieved significant improvements across all met-

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rics compared to in-dividually received CSI data, demonstrating enhanced overall data stability.

Table. 2. Comparative Analysis of Stability Metrics Between Fused Processed Data and Raw Data

Items	Standard Deviation	Entropy	ADF Test: ADF_Stat	ADF Test: P_Value	Volatility	MAE
Processed Data	5.77	3.09	-7.80	7.31E-12	5.77	4.17
Wi-Fi Receiver 1 Data	16.47	1.26	-6.98	8.44E-10	16.47	13.91
Wi-Fi Receiver 2 Data	16.80	1.34	-5.36	4.09E-06	16.80	14.52
Wi-Fi Receiver 3 Data	16.6	1.28	-5.08	1.54E-05	16.65	14.19
Wi-Fi Receiver 4 Data	13.95	0.99	-6.26	4.30E-08	13.95	10.24

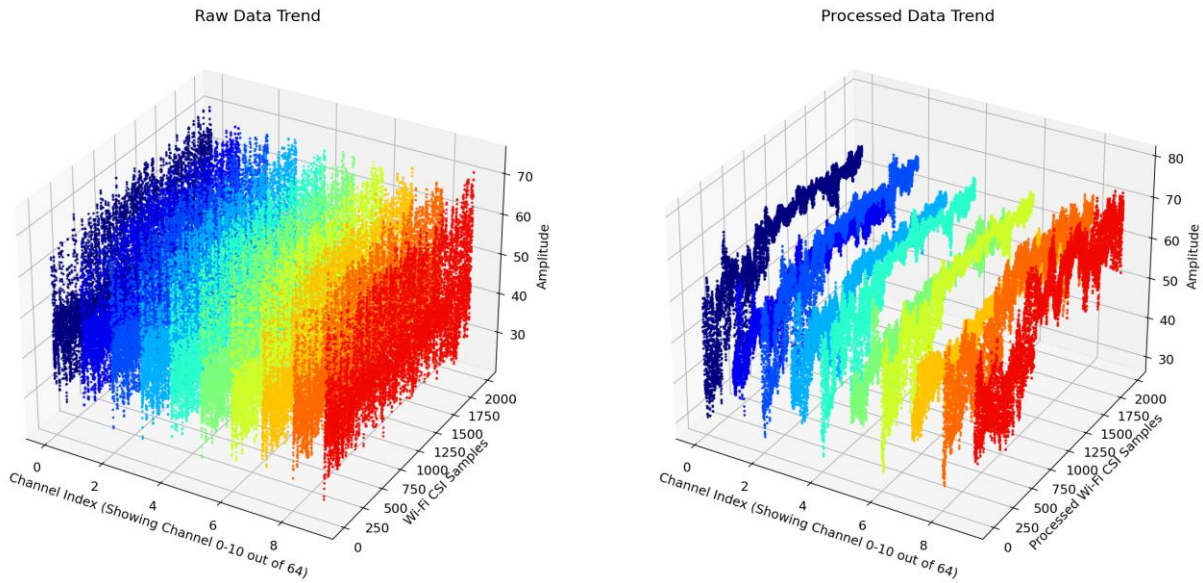

Fig. 11. Comparison of Data Before and After Processing

From Fig. 11, it is clear that the Wi-Fi CSI receiver array demonstrates significantly higher data stability. When incorporated into the model training, it contributes to more stable training results, with shorter training times as well.

LSTM is well-suited for processing sequential data and capturing long-term dependencies, aligning with the temporal characteristics of Wi-Fi CSI and the continuity of human activities. However, in NLOS environments, increased noise and signal attenuation can degrade performance. To address this, CNN is integrated for robust feature extraction, enhancing data representation before sequential processing by LSTM. Additionally, an Attention mechanism dynamically adjusts feature fusion between CNN and LSTM, improving adaptability to varying environmental conditions.

Next, we compare different models to verify the effectiveness of the proposed CNN-LSTM-Attention (C-L-A) model. We use four performance metrics: Accuracy, Precision, Recall, and F1 Score. As shown in Fig. 12, in the Line-of-Sight (LOS) scenario, the C-L-A model outperforms other models in all evaluation metrics, and its performance is comparable to that of the LSTM model in LOS conditions. In the NLOS scenario, however, the LSTM model's performance significantly deteriorates due to wall obstruction. On the other hand, the C-L-A model, by integrating CNN, LSTM, and Attention mechanisms, is able to better fuse different features and maintain robustness to signal variations, especially in complex environments, allowing it to still extract valid information. Therefore, the C-L-A model demonstrates stronger adaptability and more stable performance in the presence of signal interference and obstructions.

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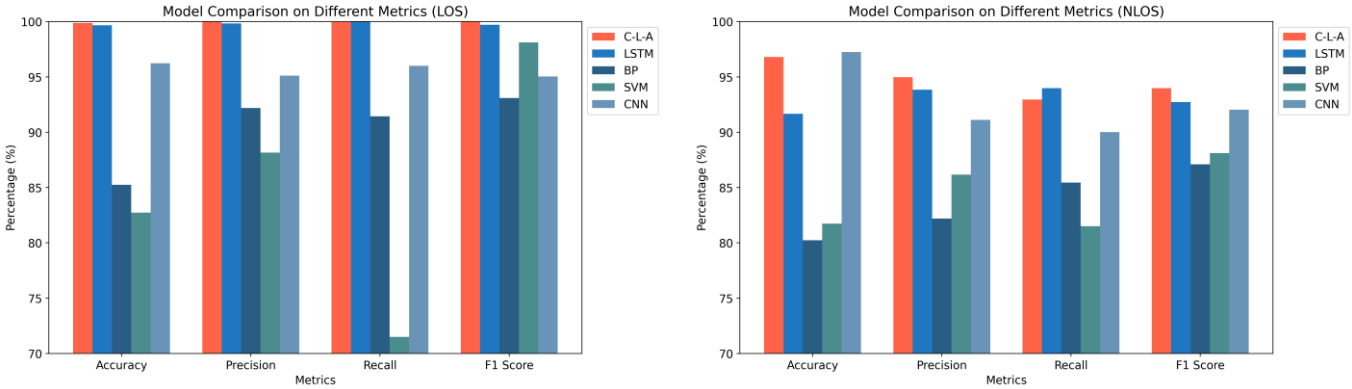

(a) Comparison of Metrics in LOS Scenario (b) Comparison of Metrics in NLOS Scenario

Fig. 12. Comparison of Model Metrics for LOS and NLOS Environments.

From the discussion of the Wi-Fi array and the analysis of the models above, it is evident that the Wi-Fi array provides reliable data support for model training. The CNN-LSTM-Attention (C-L-A) model designed in this paper maintains good performance in both LOS and NLOS environments. In summary, we deployed the C-L-A network on the STM32H7 microcontroller, implemented CSI data transmission via Bluetooth, and ultimately output the classification of movement patterns. When LOS and NLOS both datasets are labeled and trained simultaneously, as shown in Fig. 13, the training results still achieve an average accuracy of approximately 98%.

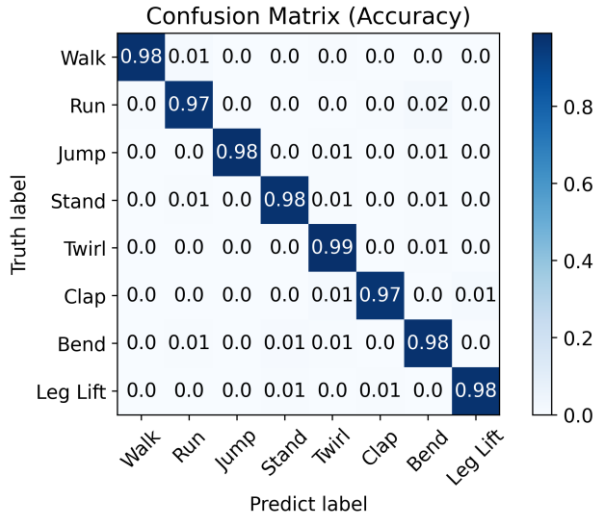

Fig. 13. The Confusion Matrix of Accuracy Generated by the C-L-A Model on STM32H7

V. Conclusion and Future Work

HAR plays a vital role in health monitoring, smart healthcare, and related fields. While vision-based methods are hindered by high deployment costs and privacy concerns, and wearable sensor solutions suffer from contact-based limitations and discomfort, Wi-Fi CSI-based approaches offer a promising non-intrusive alternative. However, existing solutions often rely on cloud-based processing and struggle in occluded or NLOS scenarios, limiting their real-world deployment.

To address these limitations, we developed a low-cost 2×2 Wi-Fi CSI array acquisition system based on the ESP32 microcontroller. The system synchronously collects multi-channel CSI data, significantly improving signal stability and reducing loss, as evidenced by improvements in statistical volatility, mean absolute error, and ADF test outcomes compared to single-receiver baselines. By applying both FFT and wavelet transforms, we captured complementary time-frequency features, enhancing the model's capacity to recognize complex motion patterns.

Furthermore, we proposed a hybrid deep learning framework deployed on the STM32H7 MCU to enable on-device edge inference. Compared with conventional models (e.g., CNN, LSTM, BP, and SVM), the C-L-A model demonstrated superior generalization, attention-driven feature refinement, and robustness, achieving up to 99% accuracy under LOS and 97% under NLOS conditions.

Overall, this study presents a complete edge-deployable Wi-Fi CSI-based HAR system, spanning data acquisition, feature extraction, and real-time recognition. Future work will focus on three key directions to further support practical applications in smart homes, elderly care, and rehabilitation scenarios: first, lightweight model optimization to reduce computational overhead; second, adaptive feature selection to enhance recognition robustness; and third, the integration of IoT transmission security mechanisms alongside energy-efficient deployment strategies—especially critical for ensuring reliable operation under low-latency and resource-constrained IoT conditions [41], [42].

Acknowledgments

The authors would like to express their sincere gratitude to the Editor and the anonymous reviewers for their insightful and constructive comments, which have significantly improved the quality of this manuscript. The authors also thank all colleagues and participants who contributed to the data collection and experimental validation of this research.

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This article has been accepted for publication in IEEE Sensors Journal. This is the author's version which has not been fully edited and content may change prior to final publication. Citation information: DOI 10.1109/JSEN.2025.3606475

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